

Survival Analysis

Alexander McLain

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Outline

What Is Survival Analysis?

Key Concepts: $S(t)$ and $h(t)$

Kaplan-Meier Curves

The Cox Proportional Hazards Model

Review

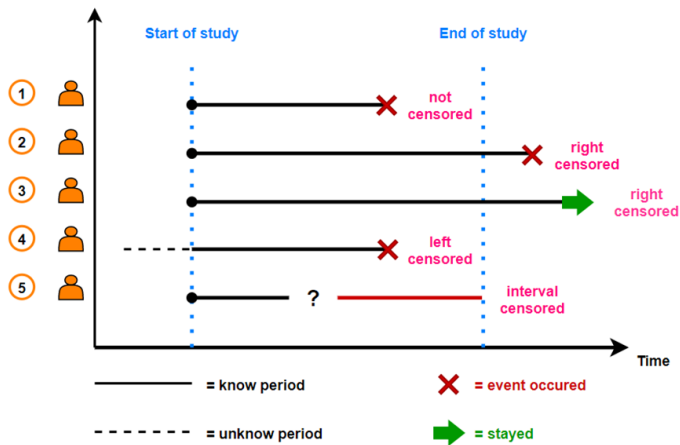
A Different Kind of Outcome

- ▶ So far we have modeled:
 - ▶ Continuous outcomes (linear regression): systolic BP, VO_2 Max.
 - ▶ Binary outcomes (logistic regression): CHD yes/no.
- ▶ Many important questions in biomedicine involve **time to an event**:
 - ▶ How long until a cancer patient dies?
 - ▶ How long until a transplant recipient rejects the organ?
 - ▶ How long until a patient experiences their first CHD event?
 - ▶ How long until a bacterium develops antibiotic resistance?
- ▶ The event could be death, disease onset, relapse, discharge, or any clearly defined transition.
- ▶ **Why not just use linear regression on the time?**

The Censoring Problem

- ▶ Survival studies have a start date and an end date. Not every participant experiences the event before the study ends.
- ▶ A participant's event time is **censored** when we do not observe it. We know only that it exceeds the censoring time.
- ▶ Common reasons for censoring:
 - ▶ The study ends before the participant has the event.
 - ▶ The participant is lost to follow-up.
 - ▶ The participant dies of a different cause.
- ▶ Censored observations are *not* missing data to be discarded. They contain real information: this person survived at least this long.
- ▶ Dropping censored observations biases results toward shorter survival times and should never be done.

Censoring



Why Not Linear or Logistic Regression?

Two naive approaches, and why both fail.

Use linear regression on the observed time:

- ▶ Censored observations are not true event times. Treating them as such systematically underestimates survival.
- ▶ Survival times are typically right-skewed and non-normal.

Use logistic regression on “did the event happen?”

- ▶ This ignores the time dimension entirely.
- ▶ A participant who died on day 10 and one who died on day 1,000 look the same to logistic regression.
- ▶ All information about *when* is lost.

Survival analysis handles both censoring and timing correctly.

The Lung Cancer Dataset

- ▶ Running example for this lecture: patients with advanced lung cancer from the North Central Cancer Treatment Group.
- ▶ Each row is one patient. Key variables:
 - ▶ time: days from study enrollment to death or censoring.
 - ▶ status: 1 = censored, 2 = dead.
 - ▶ sex: 1 = Male, 2 = Female.
 - ▶ age: age in years.
 - ▶ ph.ecog: ECOG performance score (0 = fully active, higher scores = greater impairment).
- ▶ 228 patients. Median follow-up roughly 300 days.
- ▶ About 72% of patients die during the study period; 28% are censored.

The Survival Function

- ▶ Let T be the (random) event time. The **survival function** is:

$$S(t) = P(T > t)$$

the probability of surviving beyond time t .

- ▶ Properties:
 - ▶ $S(0) = 1$: everyone is alive (event-free) at the start.
 - ▶ $S(t)$ is non-increasing: it can only stay flat or decrease.
 - ▶ $S(\infty) = 0$ in theory: eventually everyone has the event.
- ▶ $S(t)$ is a complete description of the time-to-event distribution. Estimating it from data, and comparing it between groups, is the central goal of survival analysis.

The Hazard Function

- ▶ The **hazard function** $h(t)$ is the instantaneous rate of the event at time t , given that the event has not yet occurred.
- ▶ Think of it as: given that you have survived until time t , how quickly are you “at risk” of the event in the next instant?
- ▶ $h(t)$ can take any non-negative value and can rise, fall, or change shape over time.

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- ▶ $h(t)$ can take any non-negative value and can rise, fall, or change shape over time.
- ▶ Examples of different hazard shapes:
 - ▶ **Constant:** the probability of failure does not change with time (some electronic components).
 - ▶ **Increasing:** risk grows over time (cancer, aging).
 - ▶ **Decreasing:** risk is high early and drops (post-surgery complications).
 - ▶ **Bathtub:** high early, low in the middle, high again later (human mortality).
- ▶ The Cox model (later) puts regression predictors on the hazard function.

Estimating $S(t)$: The Kaplan-Meier Estimator

- ▶ We do not know the true $S(t)$. We estimate it from data.
- ▶ The **Kaplan-Meier (KM) estimator** is the standard non-parametric method. At each event time t_j , it updates the survival estimate:

$$\hat{S}(t) = \prod_{j: t_j \leq t} \left(1 - \frac{d_j}{n_j}\right)$$

where d_j = number of events at time t_j and n_j = number at risk just before t_j .

- ▶ The KM curve is a **step function**: it stays flat between event times and drops at each event.
- ▶ Censored observations are included in the “at risk” count up to their censoring time and then are removed.
- ▶ In R : `survfit(Surv(time, status) ~ 1, data = df)`

Key Features of a KM Curve

- ▶ **Each downward step** occurs at an observed event time. The height of the step depends on how many people were at risk and how many had the event at that moment.
- ▶ **Tick marks** on the curve indicate censored observations. A censoring tick means: “this person left follow-up here, still alive (or event-free) at that point.”
- ▶ **Median survival** is read where the curve crosses 0.5. It is the time by which half the population has experienced the event.
- ▶ **Confidence bands** (shaded or dashed) widen over time as the number at risk shrinks and estimates become less certain.
- ▶ **The right tail is unreliable.** When only a few people remain at risk, each event causes a large step. Interpret carefully.

The Log-Rank Test

- ▶ When comparing two or more survival curves, we need a formal test. The **log-rank test** is the standard choice.
- ▶ Null hypothesis: the survival curves are identical across groups. Formally, $H_0 : S_1(t) = S_2(t)$ for all t .
- ▶ The log-rank test compares the *observed* number of events in each group to the *expected* number under H_0 , at every event time.
- ▶ It produces a p-value. A small p-value means the survival experience differs between groups in a way unlikely to occur by chance.

The Log-Rank Test in R

- ▶ In R : `survdifff(Surv(time, status) ~ group, data = df)`
- ▶ **Important limitations of the log-rank test:**
 - ▶ It gives a p-value, not an effect size.
 - ▶ It works best when the hazard ratio between groups is approximately constant over time.
 - ▶ It cannot adjust for confounders. The Cox model is needed for that.

Median Survival and What It Tells You

- ▶ The KM curve directly gives the **median survival time**: the time by which 50% of the group has experienced the event.
- ▶ Median survival is preferred over mean survival in time-to-event data because:
 - ▶ The mean requires knowing the full distribution, including participants who have not yet had the event.
 - ▶ Survival distributions are often right-skewed; a few long-term survivors can heavily inflate the mean.
 - ▶ The median is estimable as soon as the curve crosses 0.5, even if many participants are still censored.
- ▶ **Caution:** if fewer than half the group experiences the event, the curve never crosses 0.5 and the median is not estimable.

Limitations of KM Curves

- ▶ KM curves and the log-rank test are useful, but they have two important limitations:
 1. They cannot adjust for confounders. The observed difference in survival between groups may be due to a third variable that differs between groups.
 2. They work well for one or two categorical grouping variables. Continuous predictors require a regression model.
- ▶ **Example:** in the lung data, females survive longer than males. But females in this dataset also tend to have better performance scores. Is sex truly prognostic, or is it a proxy for performance status?
- ▶ To answer this we need a model. The standard is the **Cox proportional hazards model**.

The Cox Proportional Hazards Model

- ▶ The Cox model puts predictors directly on the hazard function:

$$h(t | \mathbf{x}) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

- ▶ $h_0(t)$: the **baseline hazard** — the hazard for a participant with all predictors equal to zero. It is left completely unspecified. This is what makes the Cox model “semi-parametric.”
- ▶ $\exp(\beta_j)$: the **hazard ratio (HR)** for a one-unit increase in X_j , holding the others constant.
- ▶ The predictors shift the baseline hazard up or down by a multiplicative factor that does not depend on time.
- ▶ In R : `coxph(Surv(time, status) ~ x1 + x2, data = df)`

The Hazard Ratio

- ▶ $HR = \exp(\beta_j)$ is the multiplicative change in hazard for a one-unit increase in X_j .
- ▶ Three reference points, same logic as the odds ratio:
 - ▶ $HR = 1 \Leftrightarrow \beta_j = 0$: no effect on survival.
 - ▶ $HR > 1 \Leftrightarrow \beta_j > 0$: higher hazard (worse survival).
 - ▶ $HR < 1 \Leftrightarrow \beta_j < 0$: lower hazard (better survival).
- ▶ **Example:** $HR = 2.0$ for a predictor means participants with that predictor have *twice* the instantaneous risk of the event at any time, compared with the reference group.
- ▶ As with odds ratios, rescale continuous HRs to meaningful units: a 10-year age increase, a 5-point performance score change, and so on.
- ▶ In R : `exp(cbind(HR = coef(model), confint(model)))`

The Proportional Hazards Assumption

- ▶ The model assumes that the hazard ratio between any two individuals is **constant over time**.
- ▶ The two hazard functions are proportional: they run parallel on the log scale at every time point.
- ▶ This is the key assumption. It fails when the effect of a predictor changes over time.
 - ▶ Example: a new drug may be highly effective early and then wear off, so the HR starts below 1 and drifts toward 1 over time.
- ▶ **How to check it:** plot the KM curves on a log-log scale. If the proportional hazards assumption holds, the curves should be approximately parallel.
- ▶ When the assumption fails, there are extensions (time-varying coefficients, stratified Cox models) but these are beyond today's scope.

Cox Model vs. Log-Rank Test vs. KM Curves

	KM Curve	Log-Rank Test	Cox Model
Visualize survival	Yes	No	No
Compare groups	Visually	Formally	Formally
Adjust for confounders	No	No	Yes
Continuous predictors	No	No	Yes
Effect size (HR)	No	No	Yes

In practice, you use all three together:

1. KM curves to visualize survival by group.
2. Log-rank test as an unadjusted group comparison.
3. Cox model for adjusted estimates and to handle continuous predictors.

Review: Survival Analysis in Three Steps

1. **Understand the data structure.** Every participant needs a time and an event indicator. Censored observations stay in the analysis.
2. **Describe survival with KM curves.** Estimate $S(t)$ non-parametrically. Compare groups with the log-rank test. Report median survival with 95% CI.
3. **Model with Cox regression.** Adjust for confounders, include continuous predictors, and report hazard ratios with confidence intervals.

Key quantities to report:

- ▶ Median survival time (with CI) for each group.
- ▶ Log-rank p-value for group comparisons.
- ▶ Adjusted HR and 95% CI for each predictor.
- ▶ Number of events and total follow-up time.